MINISTRY OF EDUCATION AND RESEARCH OF THE REPUBLIC OF MOLDOVA

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Report on laboratory work 3

The command line shell pipes and text filters

Course Operating Systems

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Purpose and execution record

The experiments were repeated on the Ubuntu x86_64 server on 9 November 2026. The xv6 tests use an isolated source copy and fresh generated disk. Figures are browser screenshots of a read-only execution-record console displaying saved SSH output; the exact text logs and source are included in the repository. Predictions were written to predictions.json before each corresponding command was run. The submission was prepared with coding assistance, and MIT xv6 is identified as upstream in the source provenance file.

Part 1 Move fast and find help

Cd and echo are Bash builtins; ls is an external command. Cd must change the current directory of the persistent shell itself. A child process could change only its own directory. In this run PATH had nine entries and which resolved ls to /usr/bin/ls. A nonexistent command returned 127. The ls option -t sorts by modification time, newest first. Man, man -k, ls --help and help cd supplied local help.



Figure 1.1 – Tab completion and reverse history search



Figure 1.2 – Type and which

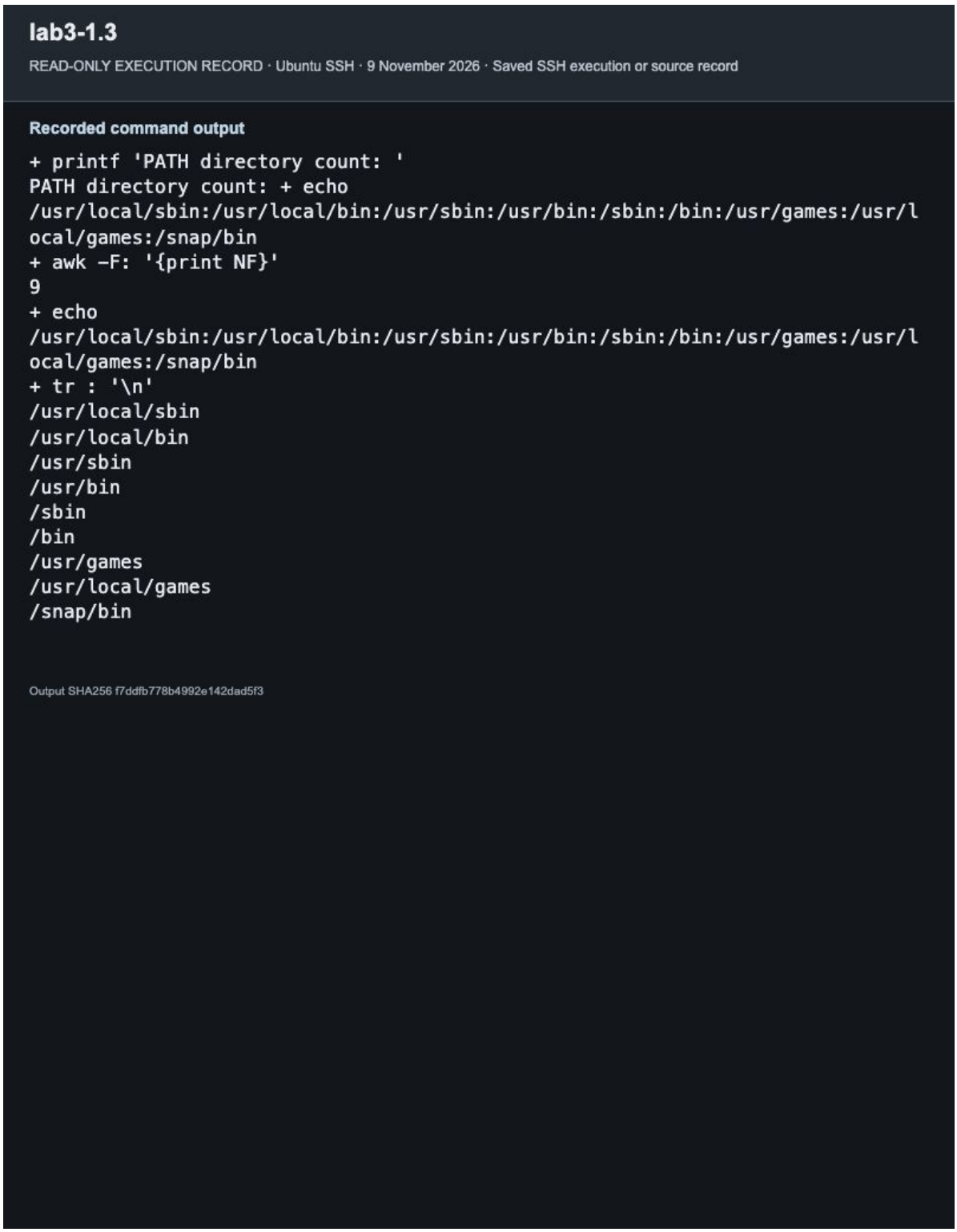


Figure 1.3 – Directories in PATH



Figure 1.4 – Exit status of a missing command

Part 2 Predictions and shell expansion

The required play directory contained a.c, b.c, file1.txt, file2.txt, notes.txt and alfa beta.txt. Each prediction below was recorded before execution; timestamps and complete stderr are retained in the JSON evidence. Differences in line layout from ls are presentation differences, not additional filenames.

Command	Prediction before run	Observed output and status	Explanation
echo *.c	a.c b.c	a.c b.c status 0	Bash expands the pattern before echo starts.
echo file?.txt	file1.txt file2.txt	file1.txt file2.txt status 0	Question mark matches one character.
echo [a-b]*	a.c alfa beta.txt b.c	a.c alfa beta.txt b.c status 0	The range matches names beginning with a or b.
echo *.xyz	literal *.xyz because no match exists	*.xyz status 0	The unmatched pattern remains literal with the default settings.
ls alfa beta.txt	two missing operands and a nonzero status	ls: cannot access 'alfa': No such file or directory ls: cannot access 'beta.txt': No such file or directory status 2	Unquoted whitespace creates two operands.
ls "alfa beta.txt"	one file named alfa beta.txt	alfa beta.txt status 0	Quotes preserve one pathname containing a space.
echo '$HOME' "$HOME" \ $HOME	$HOME /root $HOME	$HOME /root $HOME status 0	Single quotes and backslash protect the dollar sign; double quotes permit expansion.
echo {1..5} ; echo $((7 * 6))	1 2 3 4 5 then 42	1 2 3 4 5 42 status 0	Brace expansion lists values; arithmetic expansion computes 42.
echo "today is $(date +%A), I am $(whoami)"	current weekday and current login identity	today is Monday, I am root status 0	Command substitution supplies the current weekday and identity.
touch log{1..3}.txt && ls log*	log1.txt, log2.txt, log3.txt	log1.txt log2.txt log3.txt status 0	Brace expansion creates three filenames.

Echo does not discover files; Bash supplies the expanded arguments. The unquoted space in ls alfa beta.txt produces two failed lookups. Both ls "alfa beta.txt" and ls alfa\ beta.txt address the single file correctly. Single quotes retain literal text; double quotes still allow parameter and command substitution. GNU ls may display quotes for a space-containing filename when its output is a terminal, but those display quotes are not stored in the name.



Figure 2.1 – Files in the required play directory



Figure 2.2 – Globbing results



Figure 2.3 – A space containing name and the quoted fix



Figure 2.4 – Single double and escaped dollar quoting

Part 3 Redirection chaining and jobs

Out.txt received stdout, while err.txt received the diagnostic for /nope. In > all.txt 2>&1, descriptor 1 is changed first and descriptor 2 duplicates that new target. In 2>&1 > all.txt, stderr keeps the original stdout target while only stdout moves to the file. Wc -l < access.log receives stdin and therefore prints a count only; with access.log as an operand it also knows and prints the filename. A semicolon sequences regardless of the previous status; AND runs the next

command only after success; OR runs it only after failure. The background sleep was listed, terminated and reaped with status 143.



Figure 3.1 – Stdout and stderr in separate files



Figure 3.2 – Order of descriptor redirections



Figure 3.3 – Semicolon AND and OR



Figure 3.4 – Background job and termination

Part 4 Answers to the eight log questions

The artificial dataset uses the exact 2026-10-05 generator supplied in the handout. It is shared assignment input rather than real web traffic. All 500 lines were regenerated. The pipelines compare the result field, preserve tied maxima and sort their output. The four page counts sum to 500 and the seven user failure counts sum to 100. In Q8 the long listing has one extra total header line; the entries are otherwise the same.

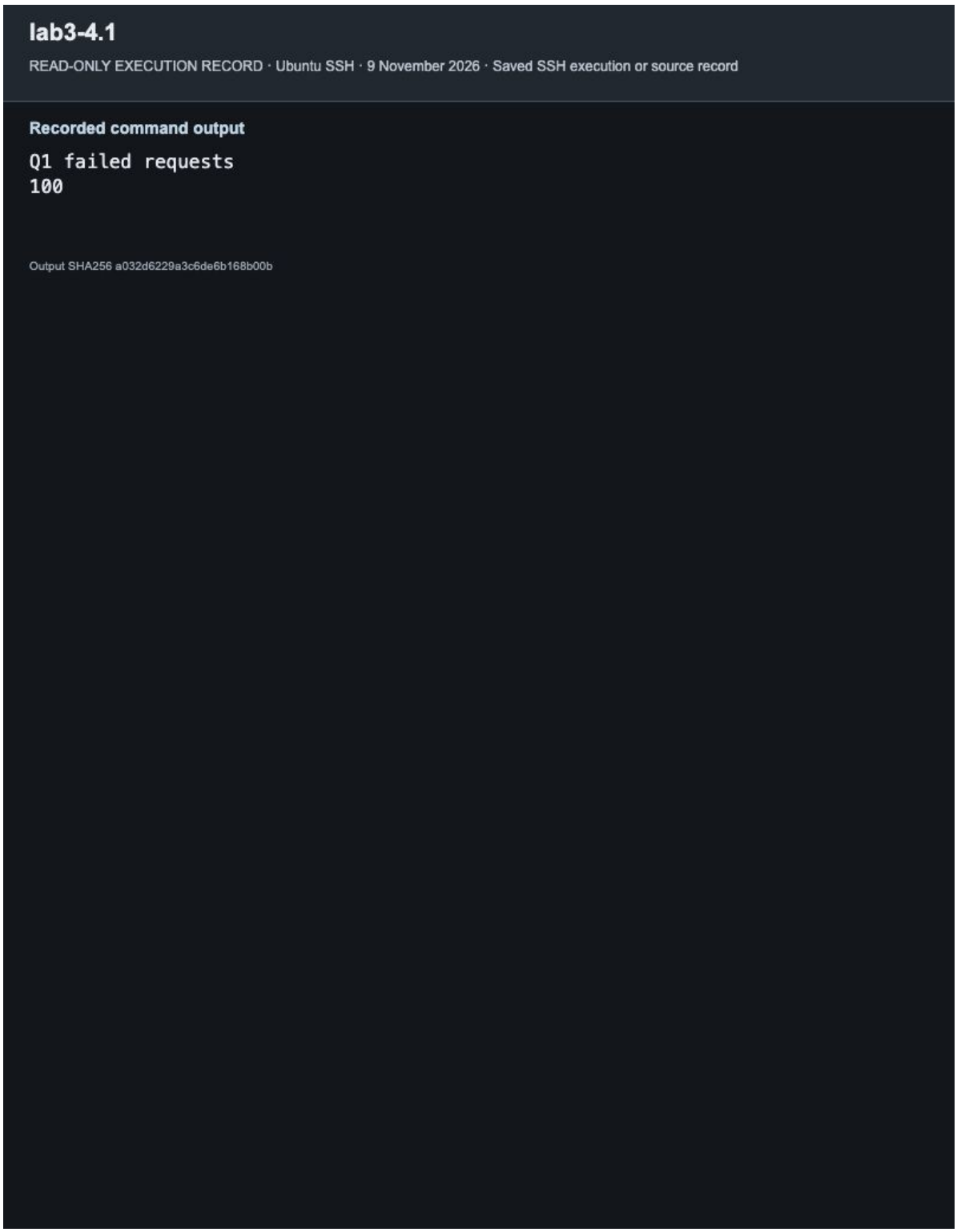


Figure 4.1 – Question 1 result



Figure 4.2 – Question 2 result



Figure 4.3 – Question 3 result



Figure 4.4 – Question 4 result



Figure 4.5 – Question 5 result



Figure 4.6 – Question 6 result

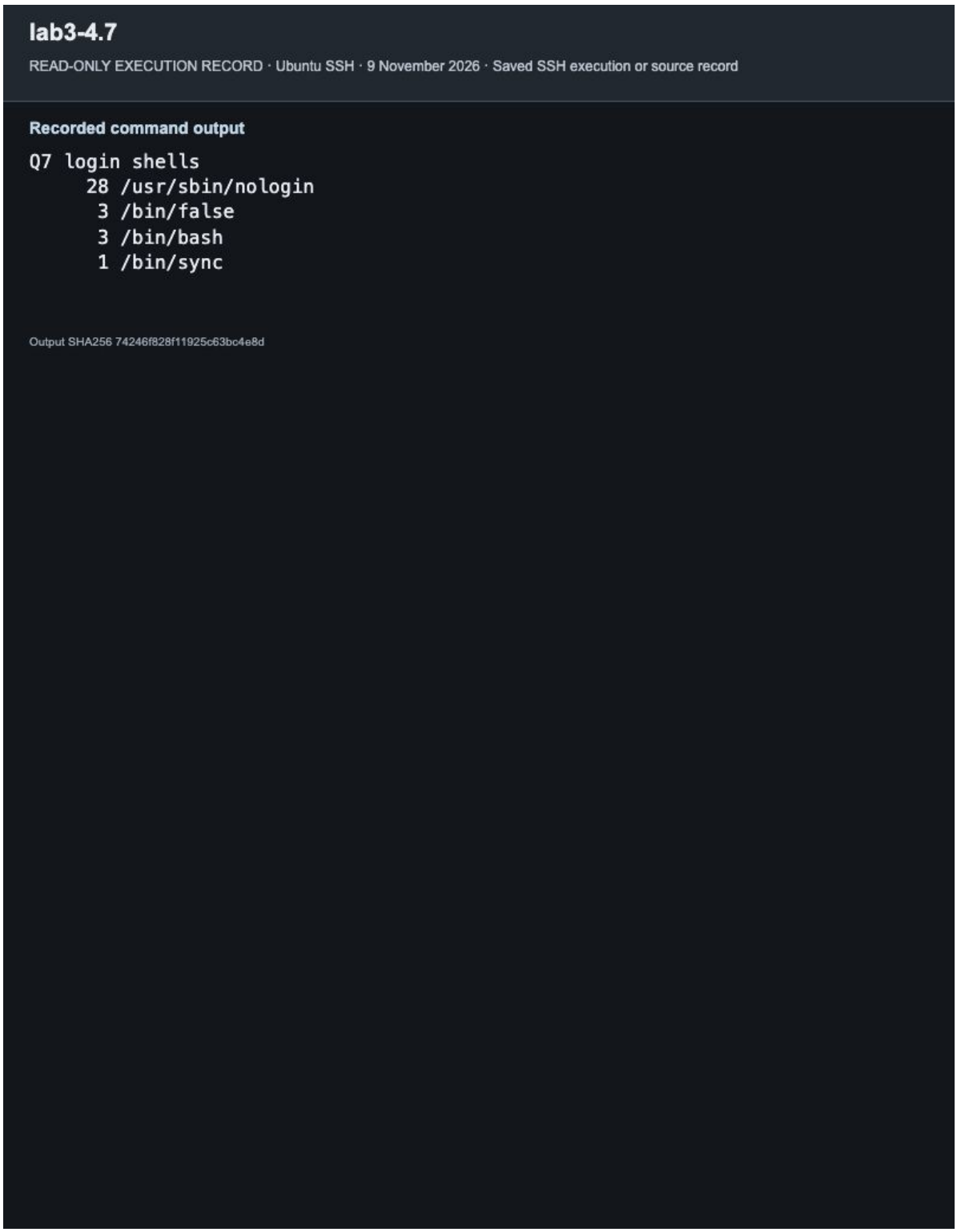


Figure 4.7 – Question 7 result

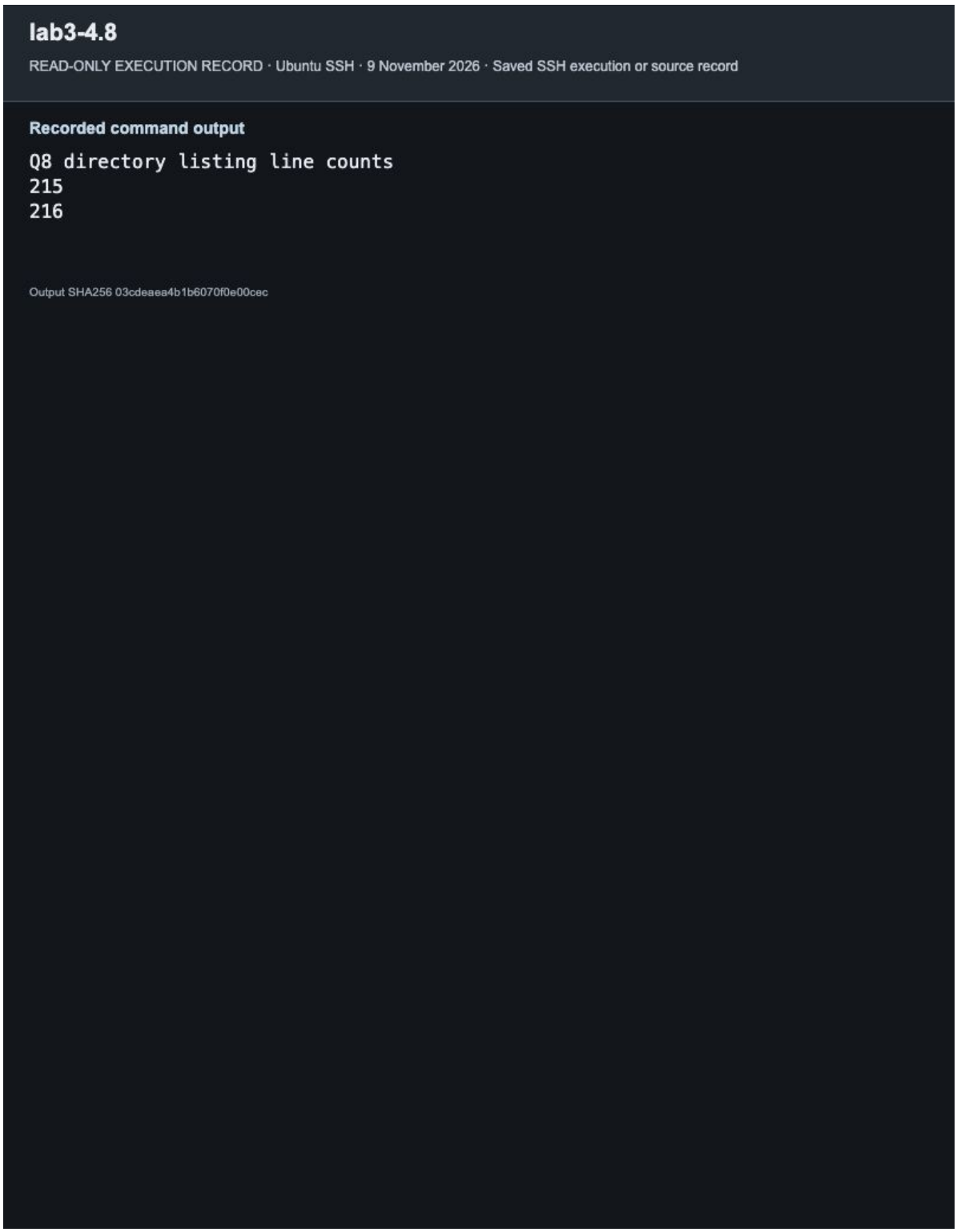


Figure 4.8 – Question 8 result



Figure 4.9 – Full run of oneliners.sh

Part 5 The report script and its error path

The shebang #!/bin/bash selects Bash when the executable script is launched directly. Before chmod +x, ./report.sh returned permission denied and status 126. The prefix ./ supplies a pathname because the current directory is not searched through PATH here. The error is written to descriptor 2 so stdout remains usable for normal results; exit 1 lets a caller detect failure. The script prints every user from user0 to user6, including zero counts, using its aggregation loop. Optional cron and bonus tasks are not claimed.

lab3-5.1

READ-ONLY EXECUTION RECORD · Ubuntu SSH · 9 November 2026 · Saved SSH execution or source record

Recorded command output

```
#!/bin/bash
if [ "$#" -ne 1 ]; then
    printf 'usage: %s readable_log\n' "$0" >&2
    exit 1
fi
if [ ! -f "$1" ] || [ ! -r "$1" ]; then
    printf 'no such file: %s\n' "$1" >&2
    exit 1
fi
awk '
  { total++ }
  $4 == "FAIL" { failed++; by_user[$3]++ }
  END {
    printf "%d requests, %d failed\n", total, failed
    for (i=0; i<7; i++) printf "user%d: %d failures\n", i, by_user["user" i]
  }
' "$1"
```

Output SHA256 d61f398259fd4e16f8e51da7

Figure 5.1 – Source of report.sh



Figure 5.2 – Successful report and execute permission check



Figure 5.3 – Missing file diagnostic and status

Conclusions

Argument expansion belongs to Bash before program execution. Descriptor duplication explains the order-sensitive redirection result, while status-dependent operators make failure handling explicit. The one-liners were verified against fixed input totals and the reusable script was tested on success and missing input. Source files, predictions and raw recordings accompany this report.